

Applied Deep Learning

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Administrative Details

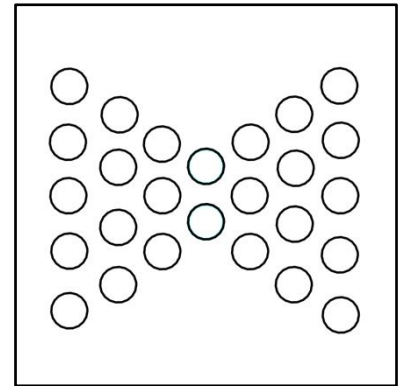
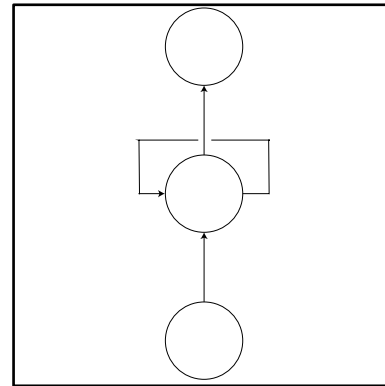
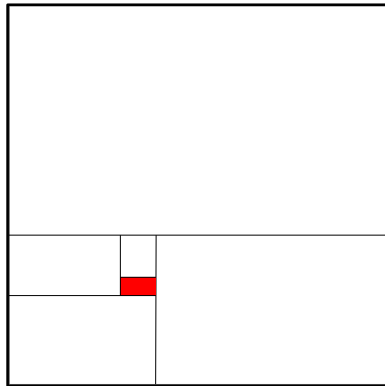
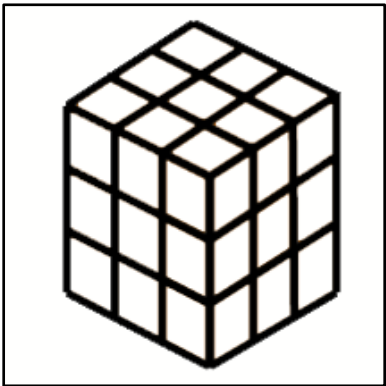
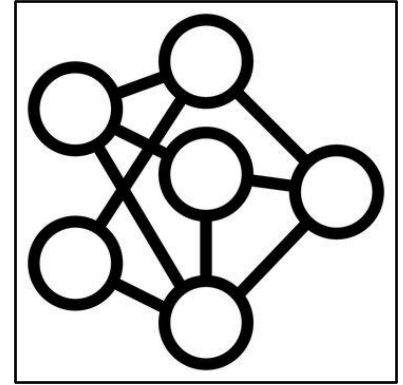
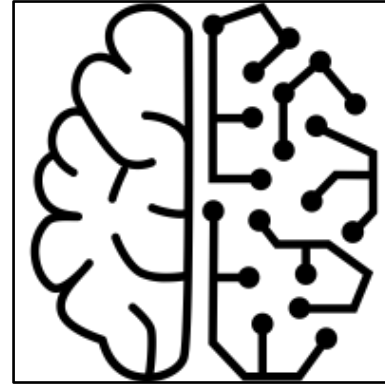
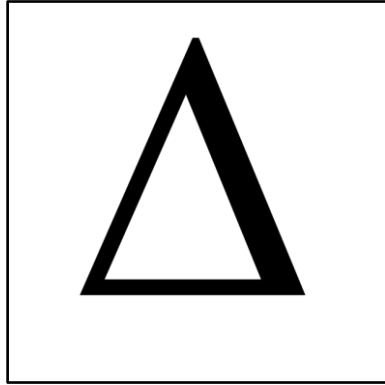
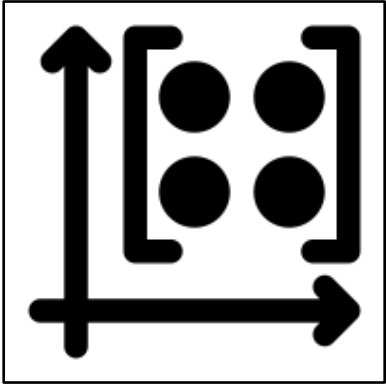


Staff Details

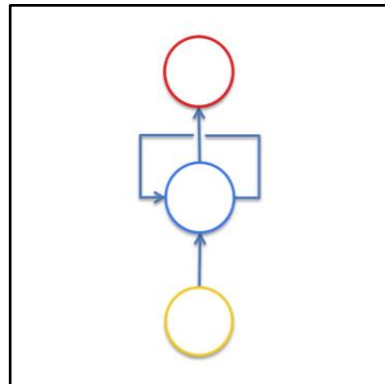
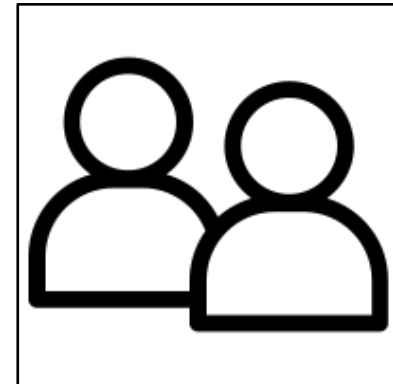
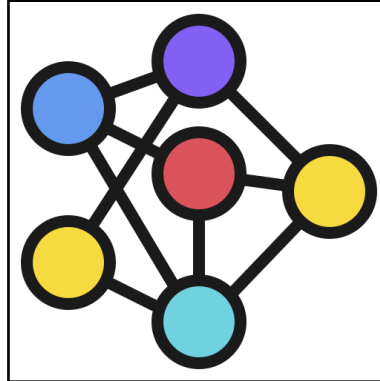
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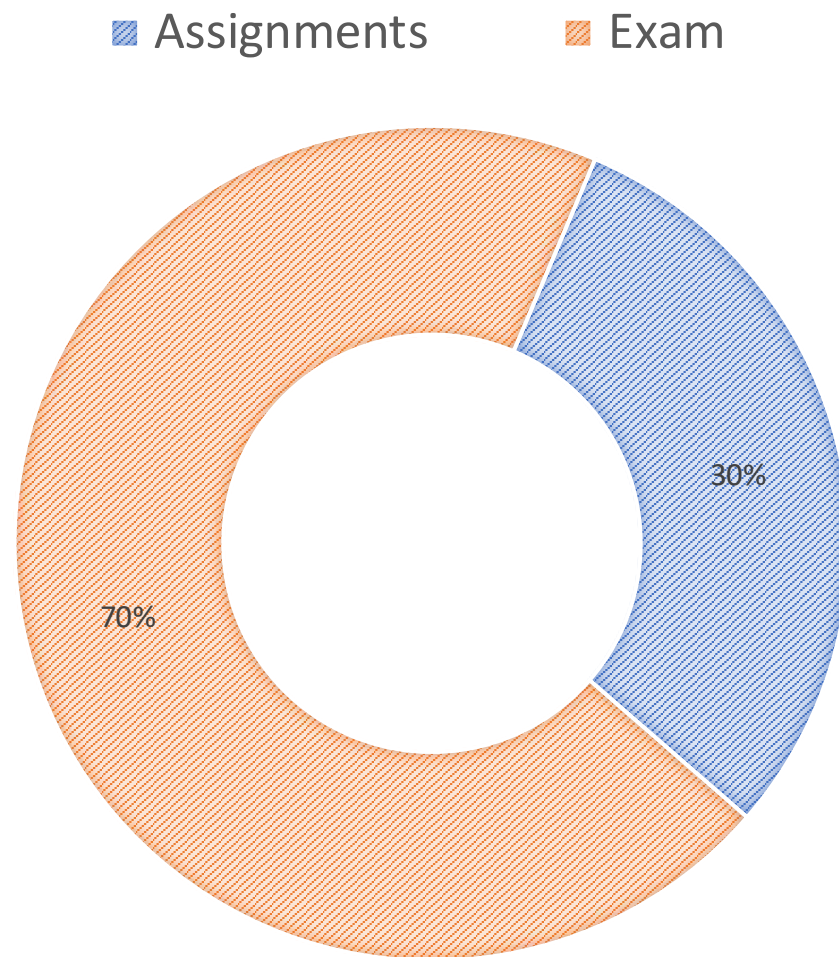
Course Plan



Home Assignments



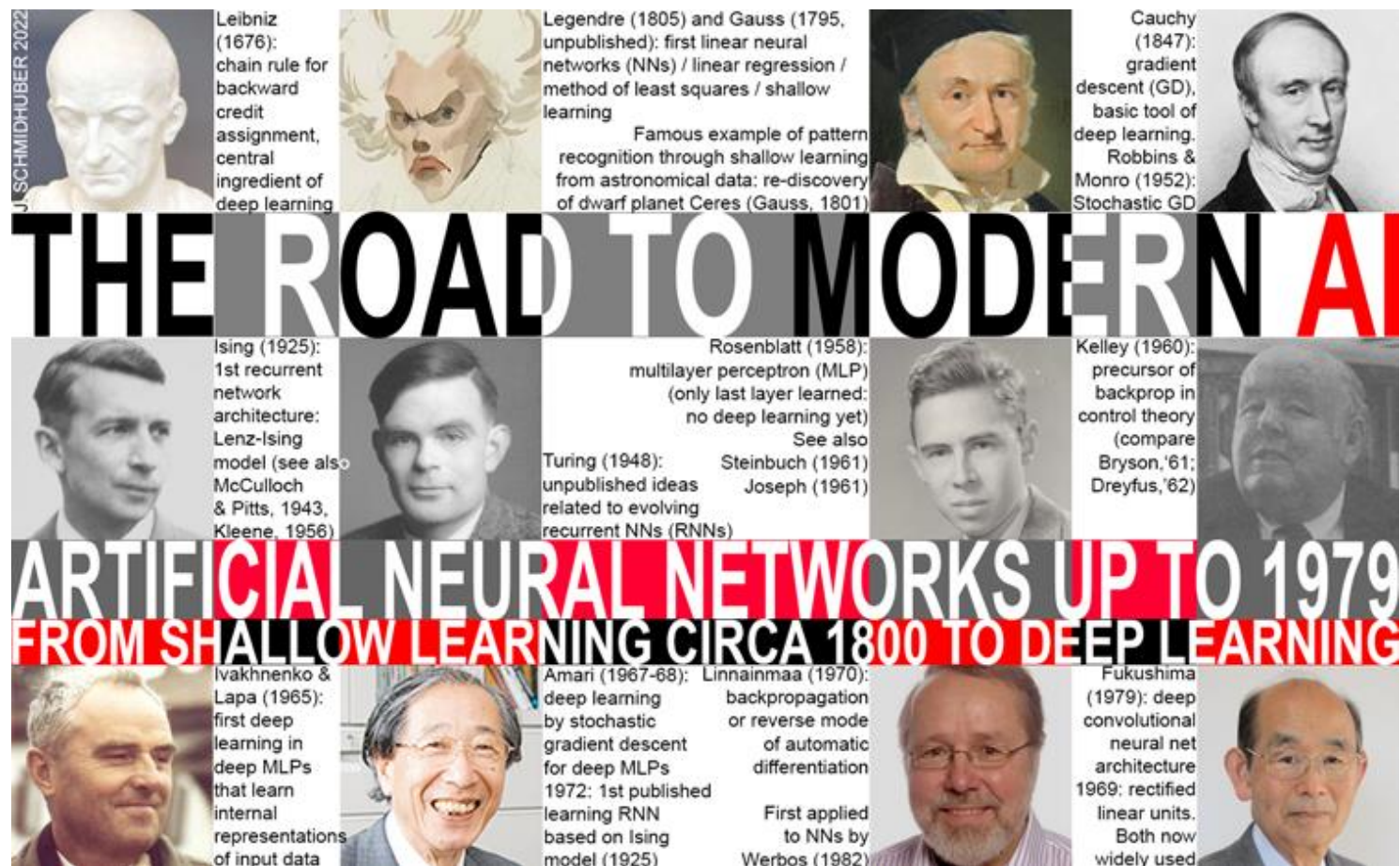
Grading Scheme



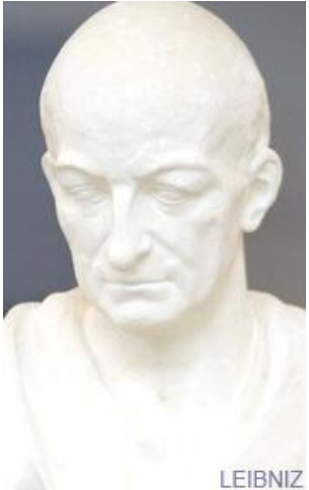
Evolution of Modern AI



Deep Learning History



1676: The chain rule



1676: Chain rule of differential calculus

allows to quantify the output of a function if its inputs are changed a bit

1847: technique of gradient descent

change weights of a neural network in the direction of the biggest local improvement



~1800: First NN / linear regression



1805: development of a linear neural network

this NN had two layers: an input layer and an output layer; the weights are adjusted s.t. the sum of squared errors is reduced

1795: Gauss made a similar development

No one called a linear NN, but rather the method of least squares (linear regression)

First pattern recognition: predict location of ceres



1920: first recurrent network



1920: the first non-learning RNN architecture

Recurrent NN have feedback connections;
this is essential for implementing a memory

1972: the Lenz-Ising architecture is
made adaptive s.t. it associates inputs
with outputs by changing its weights



1958: Multilayer feedforward NN

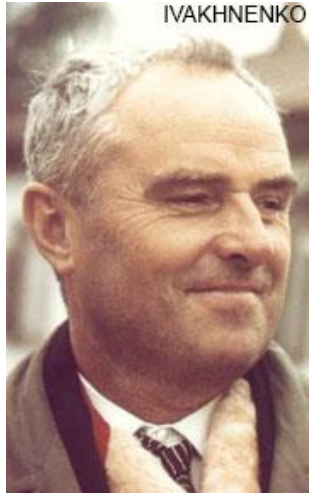


1958: Rosenblatt developed deep multilayer perceptrons (MLP)

Two layers: first not learned with random weights, and an adaptive output layer



1965: first deep learning



1965: the first deep learning algorithms were introduced;
a 1971 paper discussed 8 layer nets

Given a training set with input vectors and
output vectors, layers are incrementally
trained by regression analysis



1967: deep learning by stochastic GD



1967: Amari proposed an end-to-end training using stochastic gradient descent

They learned internal representations in a five layer MLP with two modifiable layers, trained to classify nonlinearly separable pattern classes



1970: backpropagation



1970: Linnainmaa published the backpropagation algorithm, a reverse mode of automatic differentiation

Backpropagation is essentially an efficient way of implementing Leibniz's chain rule

1960: had a precursor of backpropagation

They backpropagated derivative information through standard Jacobian matrix calculations



1978: first deep convolutional NN



1979: the neocognitron is introduced, including convolutional and downsampling layers

In 1969, Fukushima also introduced rectified linear units (ReLU)

Since 1989, LeCun's team contributed improvements for CNNs, especially for images



2011: CNN are very popular

Competition ^[GPUCNN5]	Date/Deadline	Image size	Improvement	Winner
ICDAR 2011 Chinese handwriting	May 15, 2011	variable	3.8% / 28.9%	DanNet ^[GPUCNN1-3]
IJCNN 2011 traffic signs	Aug 06, 2011	variable	68.0% (superhuman)	DanNet ^{[DAN,DAN1][R6]}
ISBI 2012 image segmentation	Mar 01, 2012	512x512	26.1%	DanNet ^[GPUCNN3a]
ICPR 2012 medical imaging	Sep 10, 2012	2048x2048x3	8.9%	DanNet ^[GPUCNN8]
ImageNet 2012	Sep 30, 2012	256x256x3	41.4%	AlexNet ^[GPUCNN4]
MICCAI 2013 Grand Challenge	Sep 08, 2013	2048x2048x3	26.5%	DanNet ^[GPUCNN8]
ImageNet 2014	Aug 18, 2014	256x256x3		VGG Net ^[GPUCNN9]
ImageNet 2015	Sep 30, 2015	256x256x3	15.8%	ResNet, ^[HW2] a Highway Net ^[HW1] with open gates



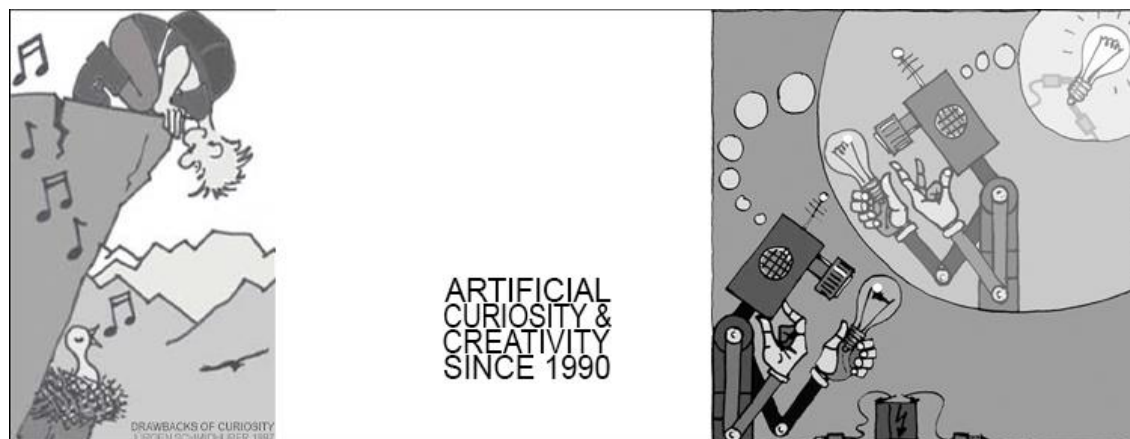
1980: graph NNs

1987: Deep learning architectures that can manipulate structured data such as graphs were proposed by Pollack

1990: Hanson introduced the stochastic delta rule, a stochastic way of training NNs by backpropagation; decades later, a version of this became popular and called dropout

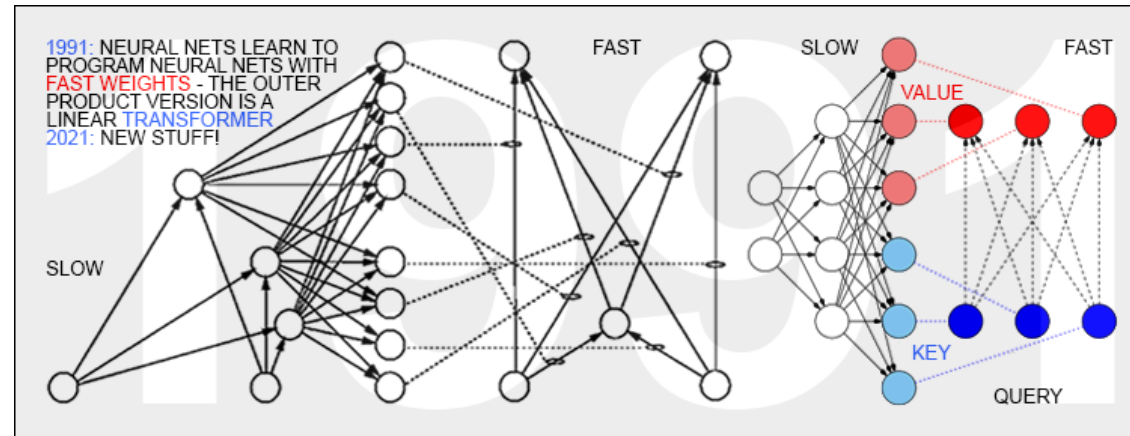


1990: generative adversarial networks



1990: two dueling NNs (a probabilistic generator and a predictor) are trying to maximize each other's loss in a minimax game

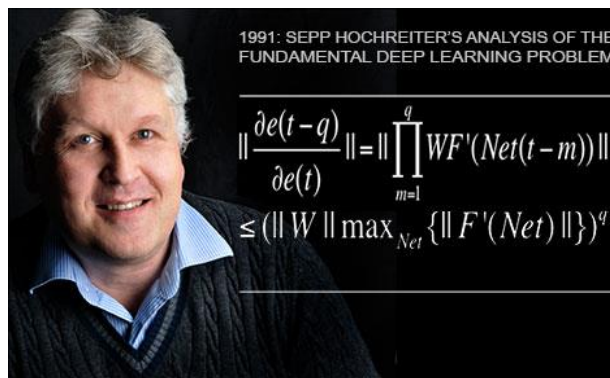
1991: transformers with linear self-attention



1991: fast weight programmers separated storage and control in an end-to-end differentiable, adaptive, fully neural way.



1991: vanishing gradients



1991: backpropagated error signals either shrink rapidly or grow out of bounds; this is called the problem of vanishing and exploding gradients

1991: long short term memory

The long short term memory recurrent neural network overcomes the fundamental problem of vanishing and exploding gradients

2006: the training method Connectionist Temporal Classification (CTC) was used for simultaneous alignment and recognition of sequences

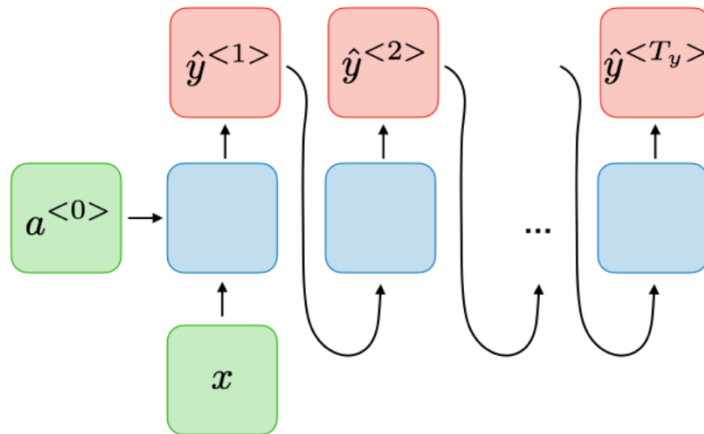


Recent Breakthroughs



Sequential Tasks

One-to-many
 $T_x = 1, T_y > 1$



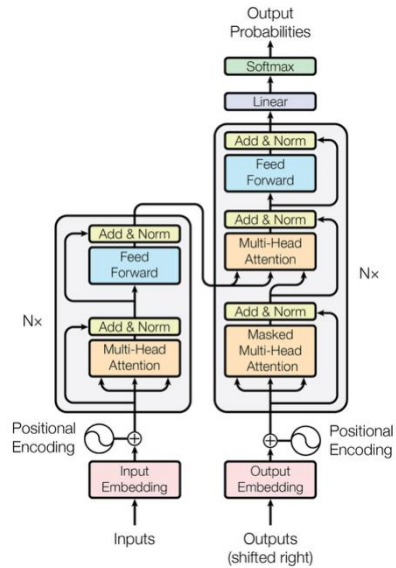
Video Generation



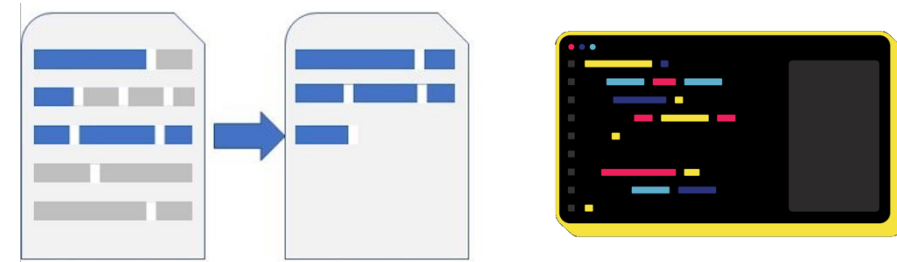
Generative Pre-trained Transformer

GPT-1

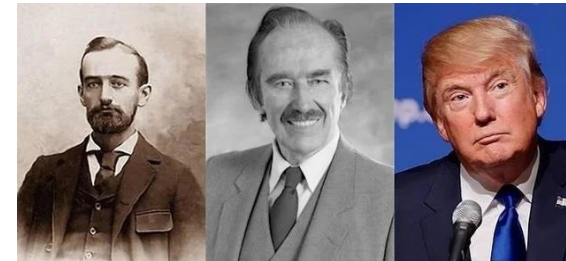
GPT-2



+



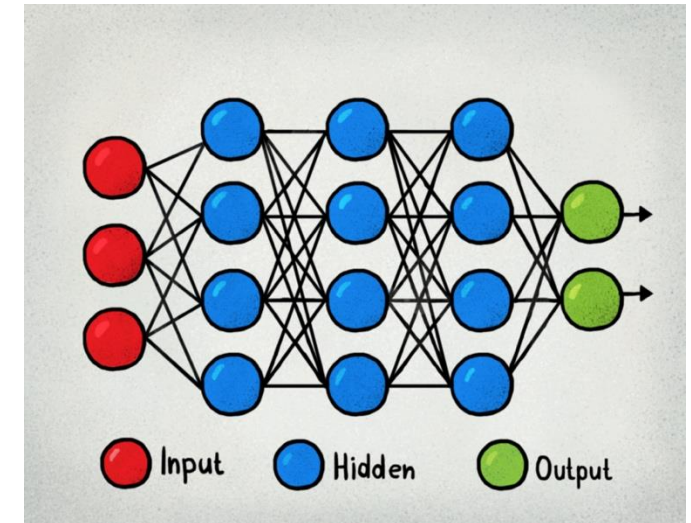
"Trump's grandfather was a pimp and tax evader;
his father a member of the KKK"



strong natural language
understanding



Generative Pre-trained Transformer 3



Language models are few-shot learners!



chatGPT



what do you think about creativity?



As an AI language model, I do not have personal opinions or emotions. However, I can tell you that creativity is often defined as the ability to generate novel and original ideas. It is a complex human trait that has been studied in various fields such as psychology, neuroscience, and philosophy. Creativity has been shown to play a major role in many aspects of human life and is often seen as a key factor in innovation, problem-solving, and artistic expression. Some researchers believe that creativity is the result of combining existing knowledge in new and unique ways, while others argue that it involves the ability to access unconscious thought processes.



chatGPT



what crime is never caught?



No crime is ever completely uncatchable, but some crimes may be more difficult to detect or prosecute than others. For example, some white-collar crimes such as financial fraud or embezzlement can be challenging to detect due to their complex nature and the need for specialized knowledge and resources to investigate. Similarly, some cybercrimes, such as hacking or identity theft, can be difficult to trace due to the anonymous and international nature of the internet.  

It's important to note that even if a crime goes undetected for a period of time, it does not mean that it cannot be solved or prosecuted in the future. Advances in technology and investigative methods are continually being made, and law enforcement agencies are working to improve their ability to detect and prosecute crimes. Ultimately, it is the goal of the criminal justice system to bring those who break the law to justice, and while some crimes may be more difficult to detect, they are never completely uncatchable.



chatGPT failure case

 When I was 6 my sister was half my age. Now I'm 70, how old is my sister?

 It is not possible to determine the age of your sister based on the information provided. You stated that when you were 6 years old, your sister was half your age, but you did not provide any information about the age of your sister at that time or her current age.  

 Yes I did. I said when I was 6 she was half my age, therefore she would have been 3

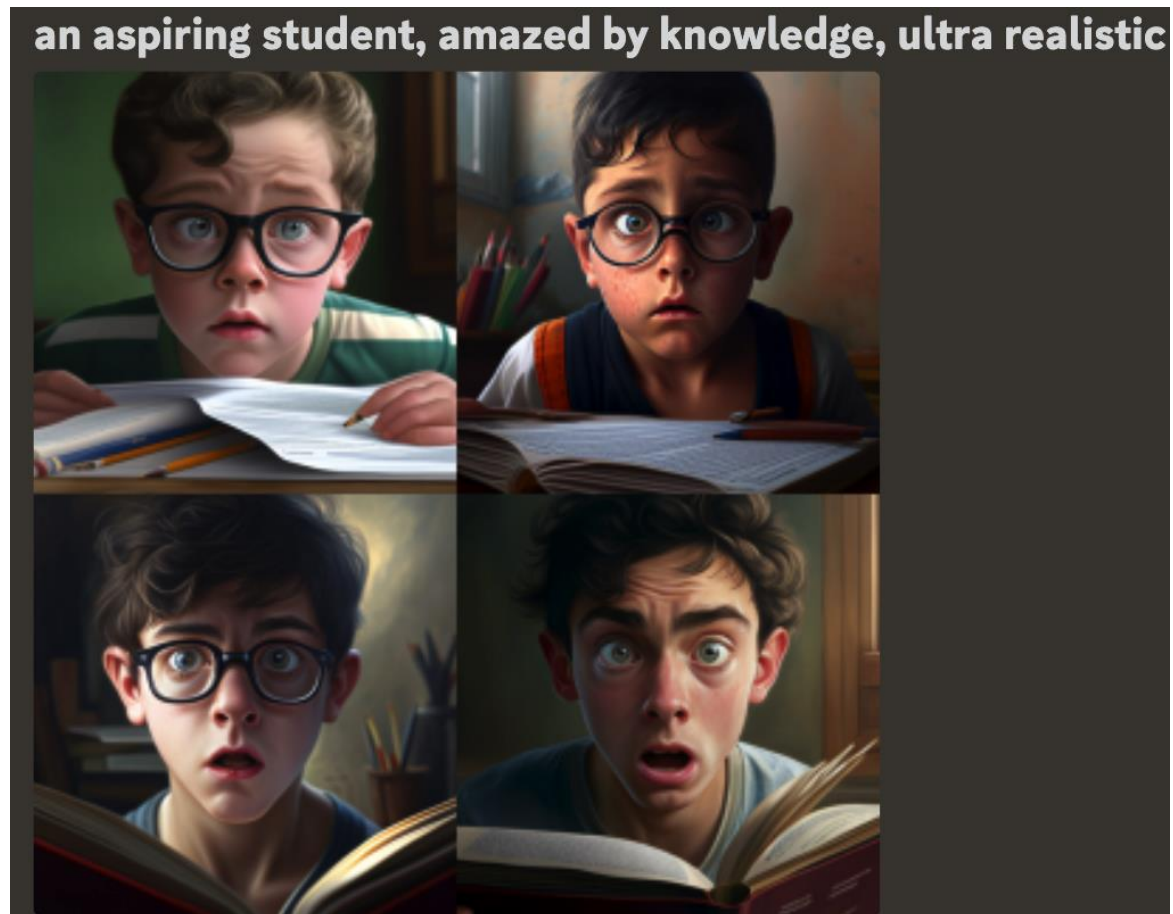
 Correct! If you were 6 years old and your sister was half your age, then she would have been 3 years old at the time. If you are now 70 years old, it's likely that your sister is also much older, but I don't have enough information to determine her current age.  

 67. The answer is 67 

 Correct! If your sister was half your age when you were 6, and you are now 70, she must be $70 / 2 = 35$ years old. Since you said she was half your age at 6, she must be $35 - 6 = 29$ years younger than you. Therefore, she must be $70 - 29 = 67$ years old.  



Midjourney



Text-to-Video Models

We're sharing our research progress early to get feedback from people outside of OpenAI and to give people a sense of what AI capabilities are on the horizon.

We will be taking several important safety steps before this research becomes available in any of our products.

Sora is a new AI model that can create realistic and imaginative scenes from text prompts.



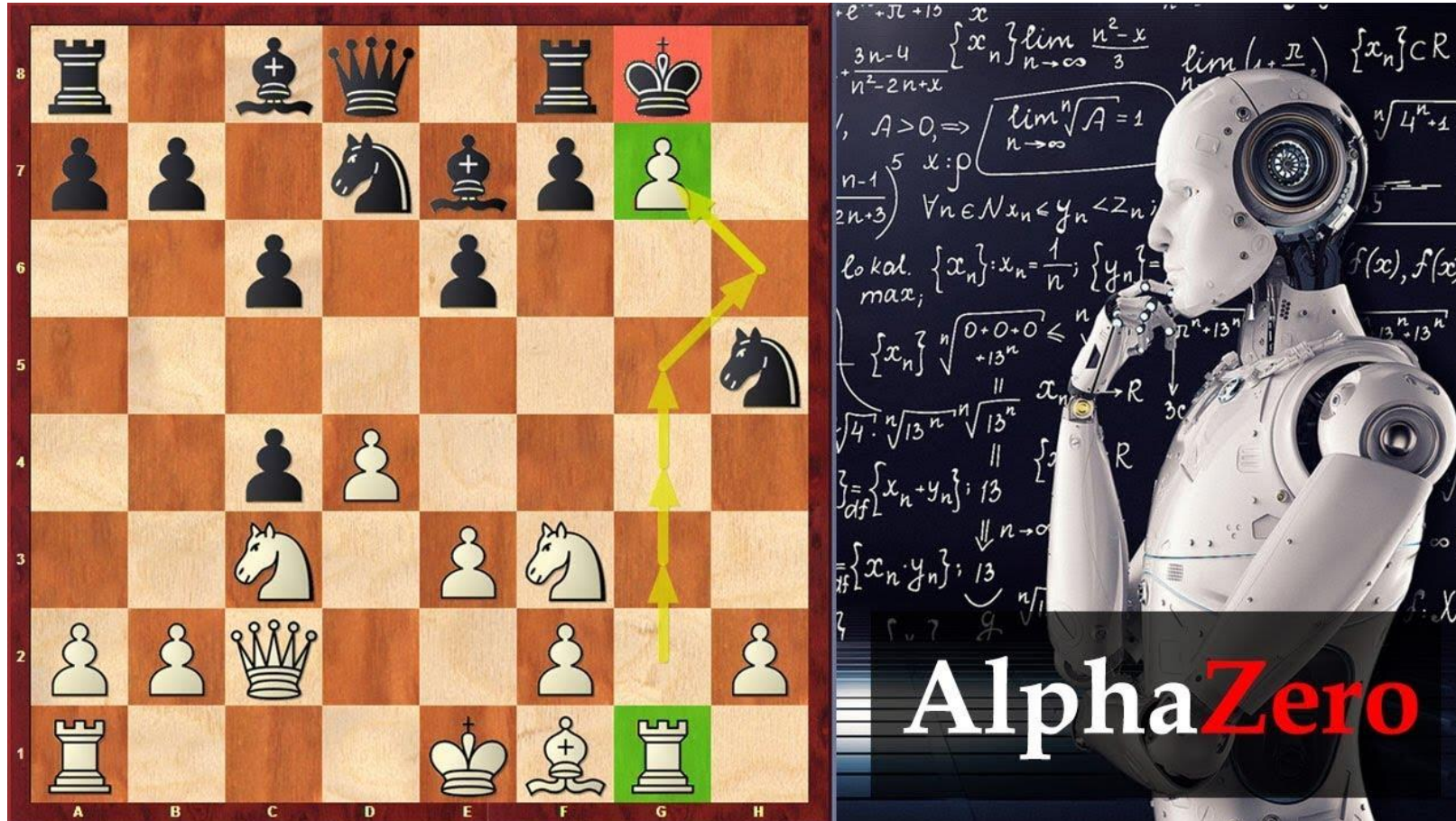
Multimodal Reasoning (Vision + Text)



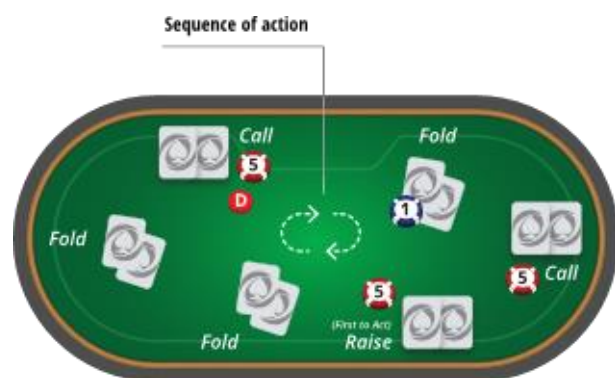
AI Coding Assistants



AI and Chess, Go, and Shogi



no-limit Texas hold'em Poker



Poker Hands

POKER.ORG POKER HAND RANKINGS			
1 ROYAL FLUSH		4 FULL HOUSE	
2 STRAIGHT FLUSH		5 FLUSH	
3 FOUR OF A KIND		6 STRAIGHT	
		7 THREE OF A KIND	
		8 TWO PAIR	
		9 PAIR	
		10 HIGH CARD	

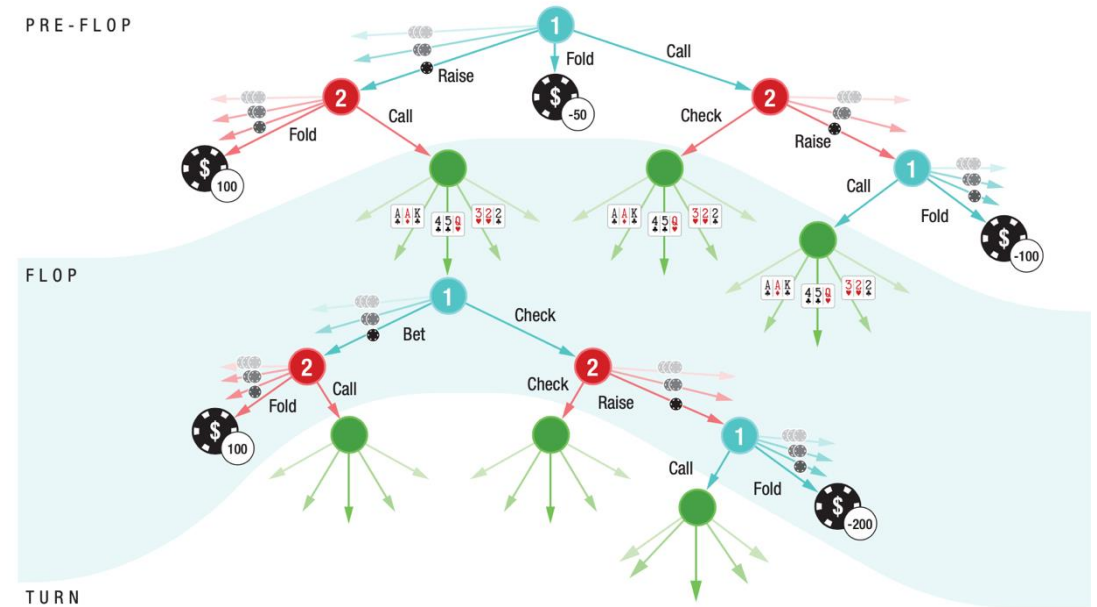


AI and Texas hold'em

DeepStack

- 44,000 hands of poker
- defeated 11 professional players
- won 49bb/100 (baseline lose 75bb/100)

Strategy



DeepStack

Score

Andrew @ThinkingPoker	0
DeepStack	0

DeepStack 16,250

Dealer 100

200

Andrew @ThinkingPoker 23,450

8 A 8 A

Hand # 22

Position bb

Player	Stack	Blind	Cards
Andrew @ThinkingPoker	23,650	200	8♦A♦
DeepStack	16,350	100	XX

Round	Actions	Cards
P		
F		XXX
T		X
R		X

Blinds 100/200

2



DeepStack

Score

Nate @NateMeyvis	1
DeepStack	1

DeepStack

17,950

200

600

600

Dealer

Nate @NateMeyvis

21,250

4 4 5 5

3

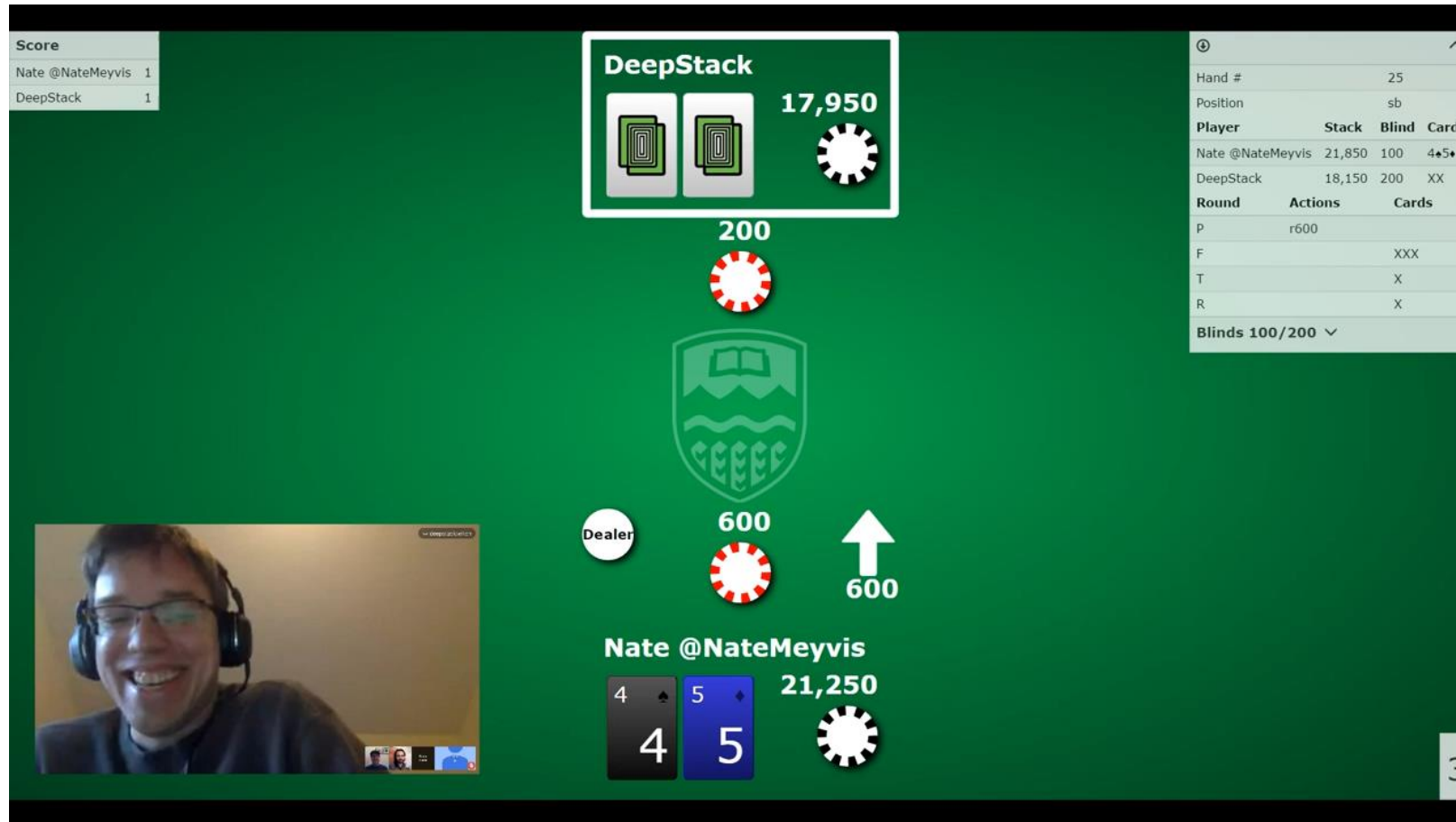
Hand # 25

Position sb

Player	Stack	Blind	Cards
Nate @NateMeyvis	21,850	100	4♠5♠
DeepStack	18,150	200	XX

Round	Actions	Cards
P	r600	
F		XXX
T		X
R		X

Blinds 100/200



Coding

